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Department of Computer Science & Engineering
(Internet of Things & Cyber Security including Blockchain Technology)

A

Mini Project Report On
"PROJECT TITLE"

Submitted in partial fulfillment of the requirement of the University of Mumbai
for the Degree
Of

Bachelor of Engineering
(Computer Science & Engineering - Internet of Things & Cyber Security including Blockchain Technology)

By

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Under the guidance of
Prof. (Guide name)

University of Mumbai
Academic Year 2025-26

Dilkap Research Institute of Engineering & Management Studies

Department of Computer Science & Engineering (Internet of Things & Cyber Security including Blockchain Technology)

Academic Year 2025-26

CERTIFICATE

This is to certify that

Group member 1, Group member 2, Group member 3, Group member 4

Of Sem IV, S.E. Computer Science & Engineering- Internet of Things

& Cyber Security including Blockchain Technology, *has satisfactorily completed the requirements of the Mini-Project* entitled

"Project title"

As prescribed by the University of Mumbai Under the guidance of

Prof.(Guide name)

Guide
(Prof.)

HOD
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Abstract

The **“Project title”** is a mobile application developed using React Native for the frontend and MongoDB and Node.js as the backend database. This app offers a platform where users can browse a wide variety of recipes and contribute their own recipes without the need for any login or registration process. The purpose of this project is to provide an easy-to-use, interactive interface for users who are passionate about cooking. The app ensures that all recipes are presented with detailed instructions, ingredients, and images, making it a comprehensive resource for cooking enthusiasts. This report outlines the app's features, architecture, implementation process, and the outcomes achieved.

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List of Abbreviations

UI	User Interface
JSON	JavaScript Object Notation

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Figure 2	React Native App Screenshot

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Table 1	Comparison of Existing Recipe Apps
Table 2	Cookbook Recipe App Features

Introduction

1.1 Introduction:

The Cookbook Recipe App is designed to cater to users who are passionate about cooking and wish to explore a variety of recipes with ease. With a simple and user-friendly interface, the app enables users to search for recipes based on ingredients or recipe names, view detailed step-by-step instructions, and add their own recipes to the collection. The app's primary focus is on providing essential recipe features without requiring users to create an account, ensuring quick access to the recipes they need. This project aims to create an intuitive platform that brings a vast collection of recipes to users' fingertips while allowing them to contribute to the growing recipe database.

1.2 Motivation:

The motivation for developing this app came from the need for a simplified recipe application that doesn't require users to sign up or log in to access basic recipe functionalities. Many existing apps require registration, making them cumbersome for users who want to quickly find or share recipes.

1.3 Problem statement & Objectives

Most recipe apps available in the market require users to register or log in before accessing their core features. This can be a barrier for users who wish to quickly find recipes or explore new dishes without going through lengthy registration processes. Additionally, many apps clutter the user experience with excessive features such as social sharing, comments, and user profiles, which can be overwhelming for those who simply want to browse recipes. The Cookbook Recipe App addresses this issue by providing a streamlined experience focused on the core functionalities: searching for recipes, viewing ingredients, and adding new recipes, without any login or registration requirements.

1.4 Organization of the Report

This report is organized as follows:

- **Chapter 2** presents the literature survey, including existing systems and identified gaps.
- **Chapter 3** describes the proposed system, including its architecture and process design.
- **Chapter 4** outlines the experiments conducted and the results obtained.
- **Chapter 5** concludes the report and suggests potential future work.

Literature Survey

2.1. Survey of Existing System

A detailed survey was conducted to analyze the existing recipe apps and identify their strengths and shortcomings. Several apps, such as Yummly, Allrecipes, and Tasty, offer comprehensive recipe databases but require user registration for saving favorite recipes or contributing new ones. The survey revealed that many users prefer apps that offer simple, unrestricted access to recipes without the need for an account. Additionally, the cluttered interfaces and multiple features found in these apps can often overwhelm users looking for a basic recipe browsing experience. This observation guided the development of the Cookbook Recipe App, focusing on providing an uncluttered, registration-free experience.

Feature	App 1: Yummly	App 2: Allrecipes	App 3: Tasty	App 4: Cookbook Recipe App (Proposed)
User Registration	Required	Required	Optional	Not Required
Recipe Contribution	Limited	Limited	Not Available	Available
Offline Access	Available (Premium)	Not Available	Not Available	Planned (Future Work)
Advertisement Free	No (Ads present)	No (Ads present)	No (Ads present)	Yes (No Ads)
Ease of Navigation	Moderate	Complex	Moderate	Simple & Intuitive
Search Filters	Advanced	Basic	Basic	Basic (Advanced planned)
Recipe Variety	Extensive	Extensive	Limited	Growing
Image & Video Support	Yes (Premium)	Yes	Yes	Yes
Personalized Profile	Yes	Yes	No	No
Login Requirement	Yes	Yes	No	No
User Rating & Reviews	Yes	Yes	No	Planned (Future Work)

2.2. Limitations of Existing Systems or Research Gap

- **Registration Requirement**: Most apps demand user login/registration, which is often seen as a hurdle.
- **Overcomplicated Features**: Many apps include complex features that clutter the interface and reduce usability.
- **Limited Offline Access**: Some existing apps do not offer offline access or have limited data when offline.

2.3. Mini Project Contribution

The Cookbook Recipe App addresses the identified gaps by:

- Providing unrestricted access to recipe search and viewing without requiring user registration.
- Offering an easy-to-navigate interface with only essential features.

Allowing users to add recipes directly to the MongoDB database, ensuring a growing collection of user-contributed content.

Proposed System

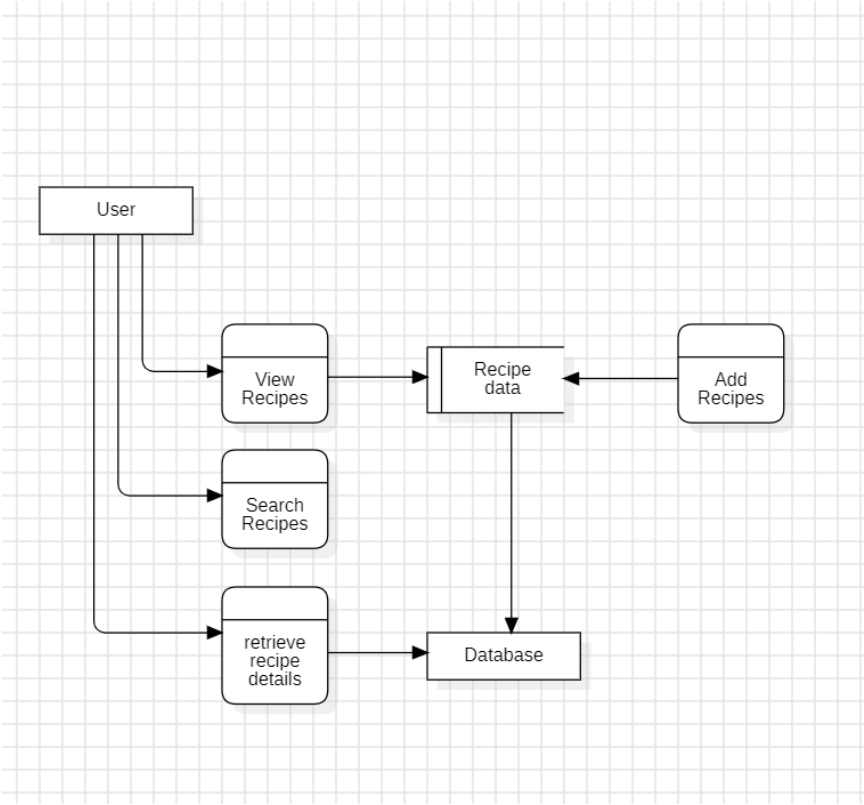
3.1 Introduction

The proposed system is a mobile application developed using React Native and MongoDB. The app allows users to search for recipes, view instructions, and contribute their own recipes. It is designed to be lightweight, responsive, and user-friendly.

3.2. Architecture/Framework

The app follows a client-server architecture, with the React Native front end acting as the client and MongoDB as the backend. The components include:

- **Front-end (React Native):** Handles the user interface and interactions.
- **Backend (MongoDB):** Manages data storage and retrieval.
- **API Layer (Node.js with Express):** Facilitates communication between the frontend and MongoDB for CRUD operations.



3.3 Algorithm and Process Design

The app employs CRUD operations to manage recipes:

1. **Search Algorithm:** The user inputs a search query, which is sent to the backend to retrieve relevant recipes from MongoDB based on keywords.
2. **Recipe Viewing:** When a recipe is selected, the app fetches the details from MongoDB, displaying the ingredients and steps.
3. **Adding Recipes:** Users can input recipe details, which are sent to MongoDB through the API.

Process Flow:

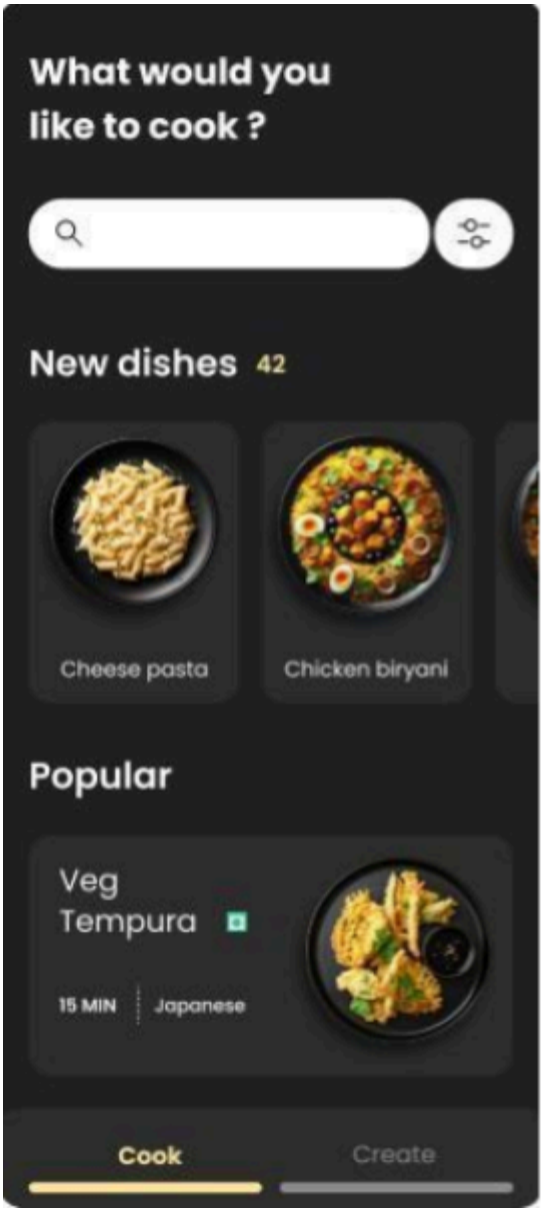
User inputs -> API call -> Database query -> Data retrieval/display

Source Code:

```
mport React, { useEffect, useState } from "react";
mport {
  Text,
  TextInput,
  View,
  Image,
  TouchableHighlight,
  Modal,
  FlatList,
  ScrollView,

 from "react-native";
mport Homestyle from "../styles/stylehome";
mport { useNavigation } from "@react-navigation/native";
mport axios from 'axios';


xport default function Homepage() {
  const [filterVisible, setFilterVisible] = useState(false);
  const navigation = useNavigation();
 const [recipies,setrecipies]=useState([])
 const[popular,setpopular]=useState([])
  const toggleFilterBox = () => {
    setFilterVisible(!filterVisible);
  };

  const timeOptions = [
    { id: "1", label: "Under 15 minutes" },
    { id: "2", label: "Under 30 minutes" },
    { id: "3", label: "Under 1 hour" },
  ];

  const Cuisines = [
    { id: "1", label: "Indian" },
    { id: "2", label: "Italian" },
```

```
const DATA = [
  { id: "1", title: "Item 1" },
  { id: "2", title: "Item 2" },
  { id: "3", title: "Item 3" },
  { id: "4", title: "Item 4" },
];
const fetchRecipes = async () => {
  try {
    const response = await axios.get('http://10.0.2.2:8080/recipes'); //
djust if using a different base URL
    console.log('Fetched recipes:', response.data); // Check if data is
eturned
    setrecipies(response.data); // Update state with fetched recipes
  } catch (error) {
    console.error('Error fetching recipes:', error.response ?
rror.response.data : error.message);
  }
};
const fetchpopular = async () => {
  try {
    const response = await axios.get('http://10.0.2.2:8080/popularrecipes')
/ Adjust if using a different base URL
    console.log('Fetched recipes:', response.data); // Check if data is
eturned
    setpopular(response.data); // Update state with fetched recipes
  } catch (error) {
    console.error('Error fetching recipes:', error.response ?
rror.response.data : error.message);
  }
};
useEffect(() => {
```

```
mport React, { useState, useEffect } from "react";
mport {
  Image,
  Text,
  TouchableOpacity,
  View,
  PermissionsAndroid,
  Platform,
  TextInput,
  ScrollView,
  FlatList,
  TouchableHighlight,
} from "react-native";
mport { launchImageLibrary } from "react-native-image-picker";
mport Createstyle from "../styles/stylecreate"; // Ensure you have this file
reated with the styles
mport axios from 'axios';

xport default function Createform({ navigation }) {
  const [imageUri, setImageUri] = useState(null);
  const [recipeName, setRecipeName] = useState('');
  const [ingredients, setIngredients] = useState('');
  const [ingredientsData, setIngredientsData] = useState([]);
  const [steps, setSteps] = useState('');
  const [stepsData, setStepsData] = useState([]);
  const [tags, setTags] = useState('');
  const [tagsData, setTagsData] = useState([]);

  const requestStoragePermission = async () => {
    if (Platform.OS === 'android') {
      try {
        const granted = await PermissionsAndroid.request(
          PermissionsAndroid.PERMISSIONS.READ_EXTERNAL_STORAGE,
          {
```


What would you like to cook ?
New dishes 42
Cheese pasta
Chicken biryani
Popular
Veg Tempura
15 MIN
Japanese
Cook
Create

Cook Book
It's Cooking time !!
Get started

3.4. Details of Hardware and Software:

Hardware Requirements:

- A computer with at least 4GB RAM for development
- Android device/emulator for testing

Software Requirements:

- **React Native**: For building the app's user interface
- **Node.js**: For setting up the backend server
- **MongoDB:** Database management system.
- **Visual Studio Code**: IDE for coding
- **Axios**: For handling API requests

4.Experiment and Results

During the development phase, the app was tested for functionality, responsiveness, and performance:

- **Functionality Testing**: Verified that all modules (search, view, and add recipe) functioned correctly.
- **Usability Testing**: Ensured the user interface was intuitive and easy to navigate.
- **Performance Testing**: Evaluated the app's performance with a large dataset to ensure efficient data retrieval from the MongoDB database.

Results:

- The search functionality provided accurate results based on the entered keywords.
- Recipe details were displayed correctly, and adding new recipes to the database was successful.
- The app performed efficiently without noticeable lag, even with increased data entries.

Feature	Description
User Interface (UI)	Simple, intuitive, and easy to navigate
Recipe Contribution	Allows users to add their own recipes
Search Functionality	Basic search with recipe name and ingredient filters
Recipe Details	Includes ingredients, preparation steps, and cooking time
No Registration/Login	Users can access and contribute without an account
MySQL Database Integration	Stores all recipe data in a structured way
React Native Compatibility	Cross-platform app experience for Android & iOS
No Advertisements	Ad-free recipe browsing experience
Offline Access (Planned)	Future feature for accessing saved recipes offline
Future Enhancements	Adding user ratings, reviews, and recipe recommendations

5. Conclusion and Future Work

The Cookbook Recipe App successfully meets its objectives by providing a user-friendly platform for searching, viewing, and adding recipes without requiring login or registration. The app demonstrates effective use of React Native for front-end development and MongoDB for backend data management.

Future Work:

- **Offline Access**: Implementing offline storage for recipes to allow access without an internet connection.
- **Enhanced Search**: Integrating advanced search filters, such as cuisine type or preparation time.
- **Recipe Rating**: Allowing users to rate recipes, helping others find the most popular dishes.